Lecture for 10th week

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Statically Indeterminate Problems

1.General Considerations

- If the equations of statics suffice (足够) for determining the external reactions as well as the internal stress resultants, a structural system is statically determinate. Otherwise, it is statically indeterminate.

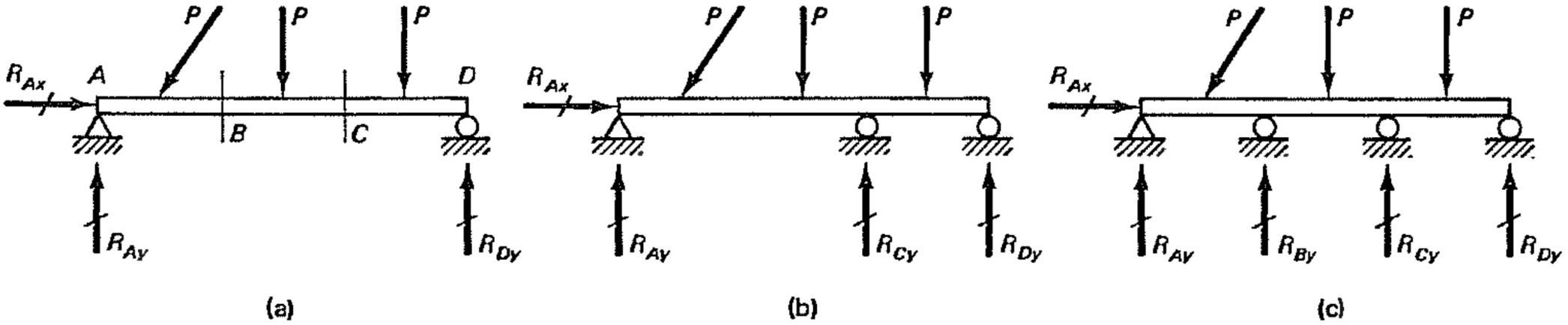


Fig. 1-16 Identical beam with identical loading having different support conditions

1.General Considerations

- The reactions that can be removed leaving a stable system statically determinate are redundant. Depending on the number (n) of the redundant internal forces or reactions, the system is said to be indeterminate to the n-th degree.

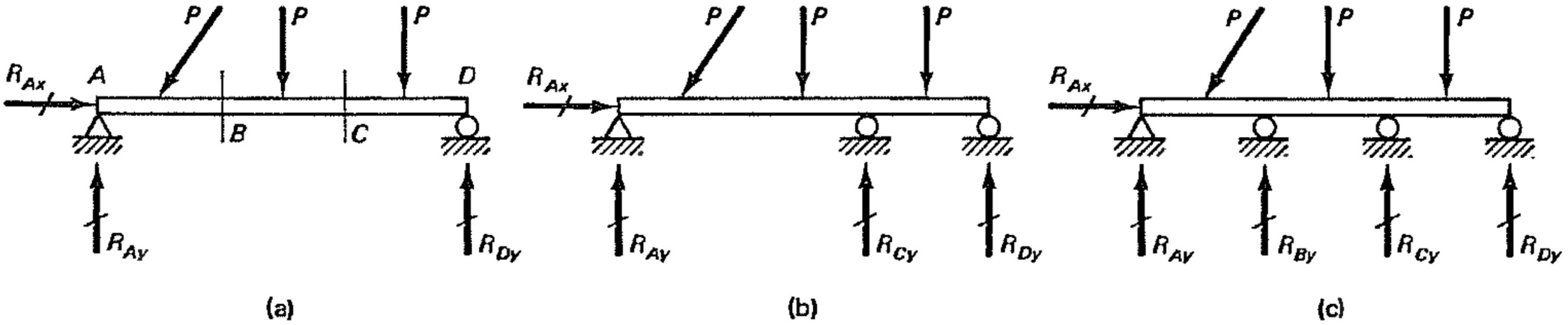
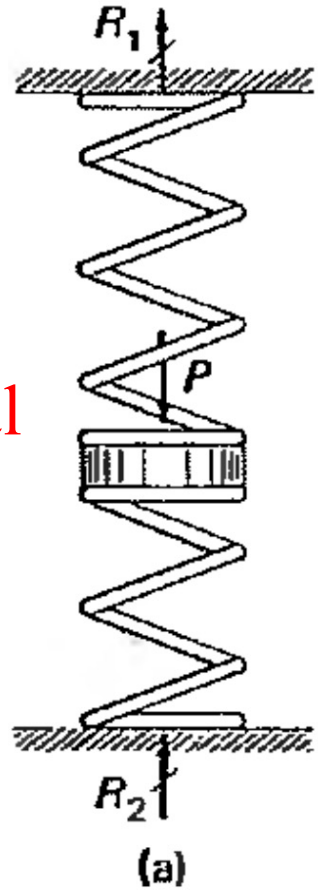


Fig. 1-16 Identical beam with identical loading having different support conditions

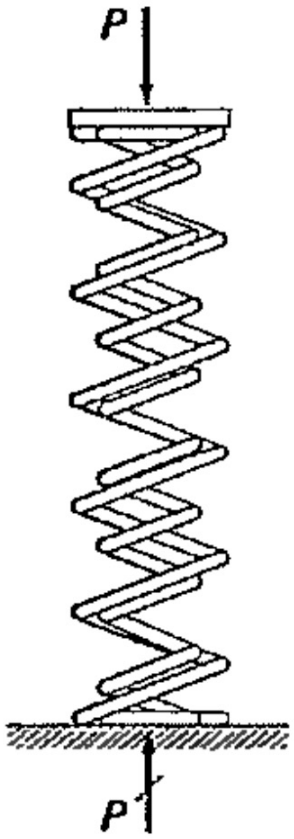
1.General Considerations

external statical indeterminacy



(a)

internal statical indeterminacy



(b)

For both instances, the deformation characteristics of the system components must be considered.

1.General Considerations

- Reduce a problem to statical determinacy such that the internal forces can readily be found. Common to all these procedures, the same three basic concepts encountered before are applied, and must be satisfied:

1. Equilibrium conditions for the system must be assured both in the local and global sense.
2. Geometric compatibility among the deformed parts of a body and at the boundaries must be satisfied.
3. Constitutive relations (stress-strain relations) for the materials of the system must be complied with.

1. General Considerations

- **Force Method**: First removing and then restoring redundant reactions such that the compatibility condition at boundaries is satisfied. Solution is obtained directly for the unknown reaction forces.
- **Displacement Method**: The compatibility of displacements of adjoining members and at the boundaries is maintained throughout the loading process, and solution for displacements are obtained from equilibrium equations.

2. Force Method of Analysis

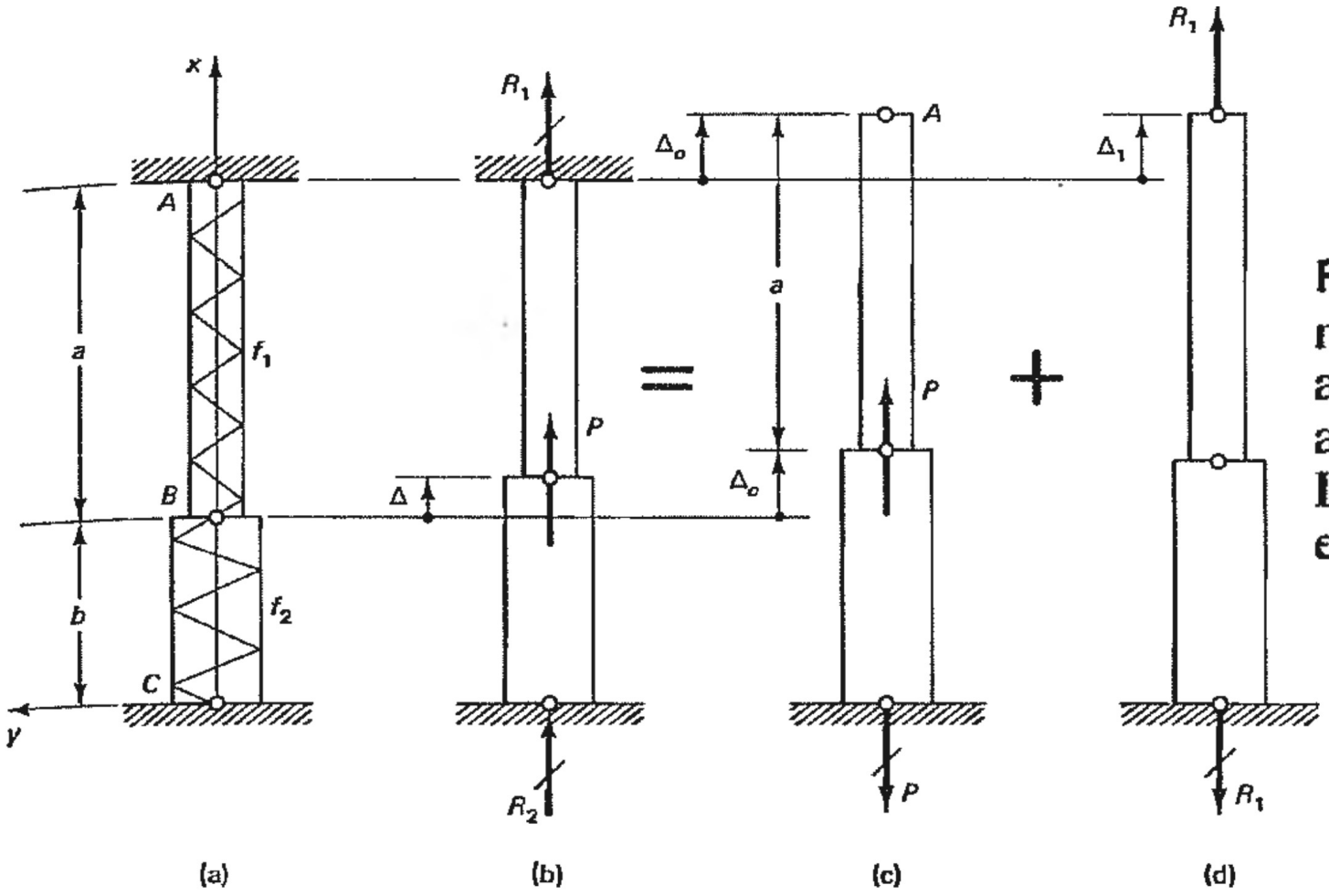
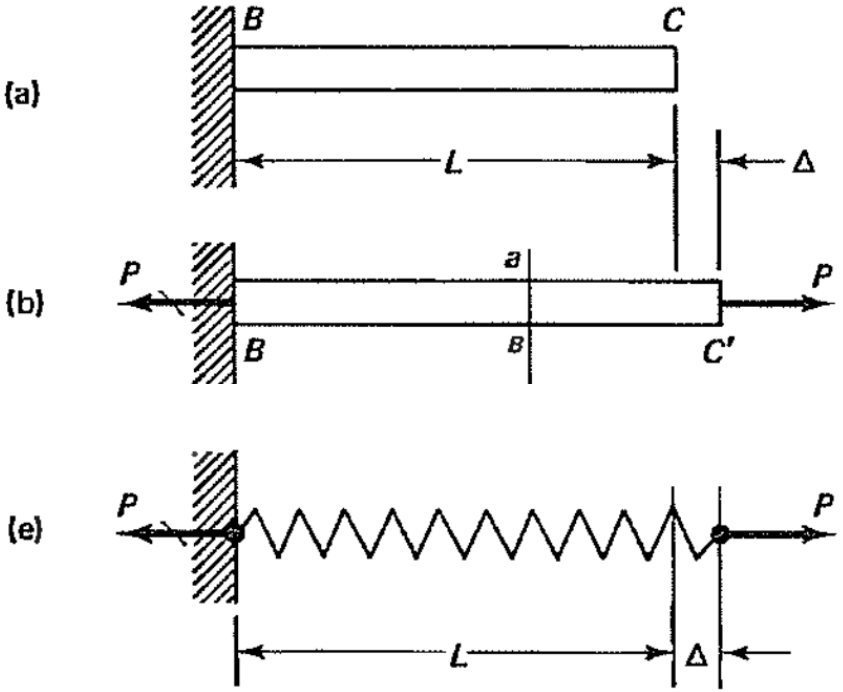


Fig. 2-43 Force (flexibility) method of elastic analysis for a statically indeterminate axially loaded bar. Deformations are greatly exaggerated.

2. Force Method of Analysis



in analogy with spring

$$\Delta = \int_A^B \frac{P_x\,dx}{A_x E} = \frac{P}{AE}\int_0^L dx = \frac{P}{AE}\left| x \right|_0^L = \frac{PL}{AE}$$

$$\Delta = \frac{PL}{AE}$$

$$\kappa = \frac{AE}{L} = \frac{P}{\Delta}$$

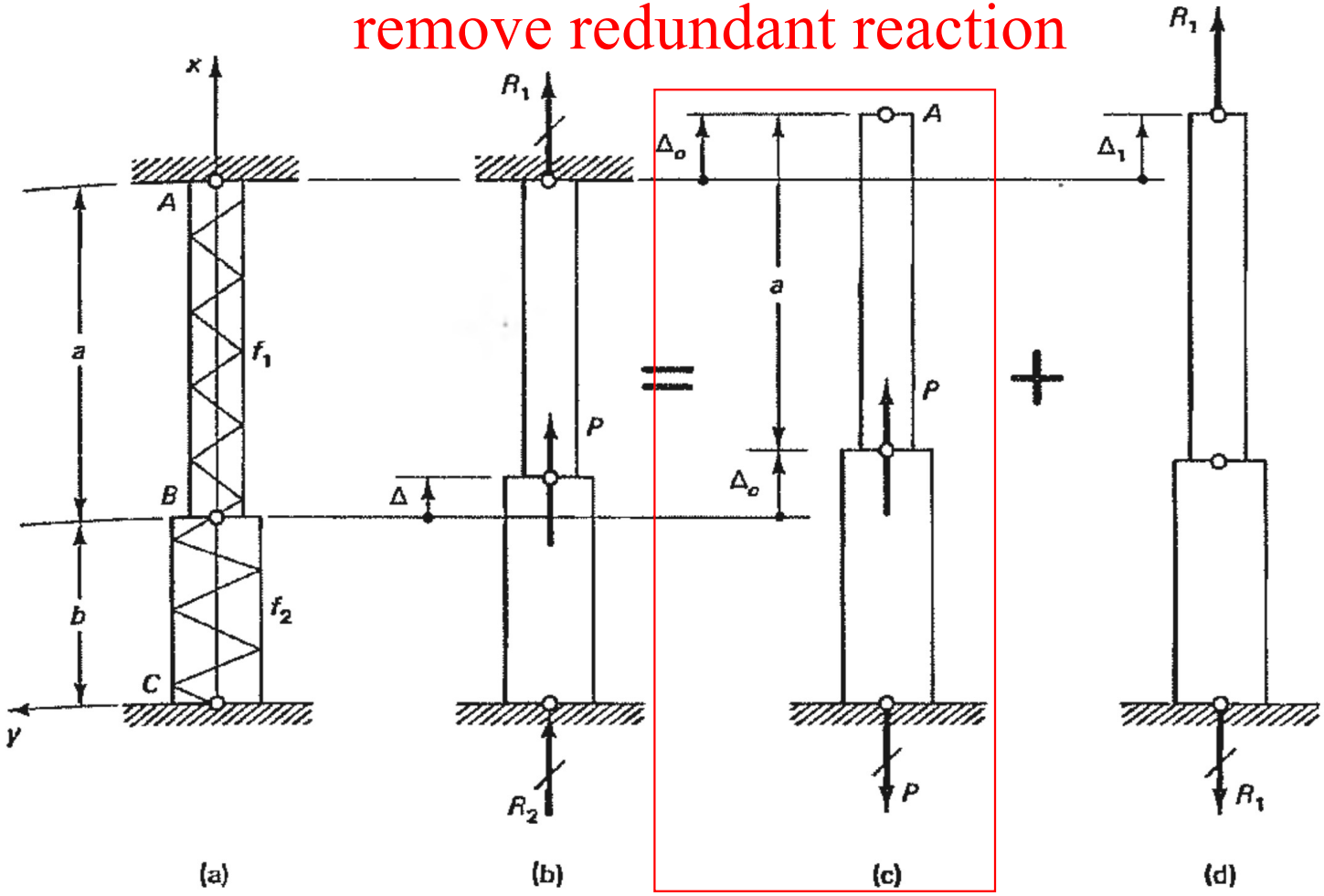
stiffness

$$f = \frac{1}{\kappa} = \frac{\Delta}{P} = \frac{L}{AE}$$

flexibility

2. Force Method of Analysis - example 1

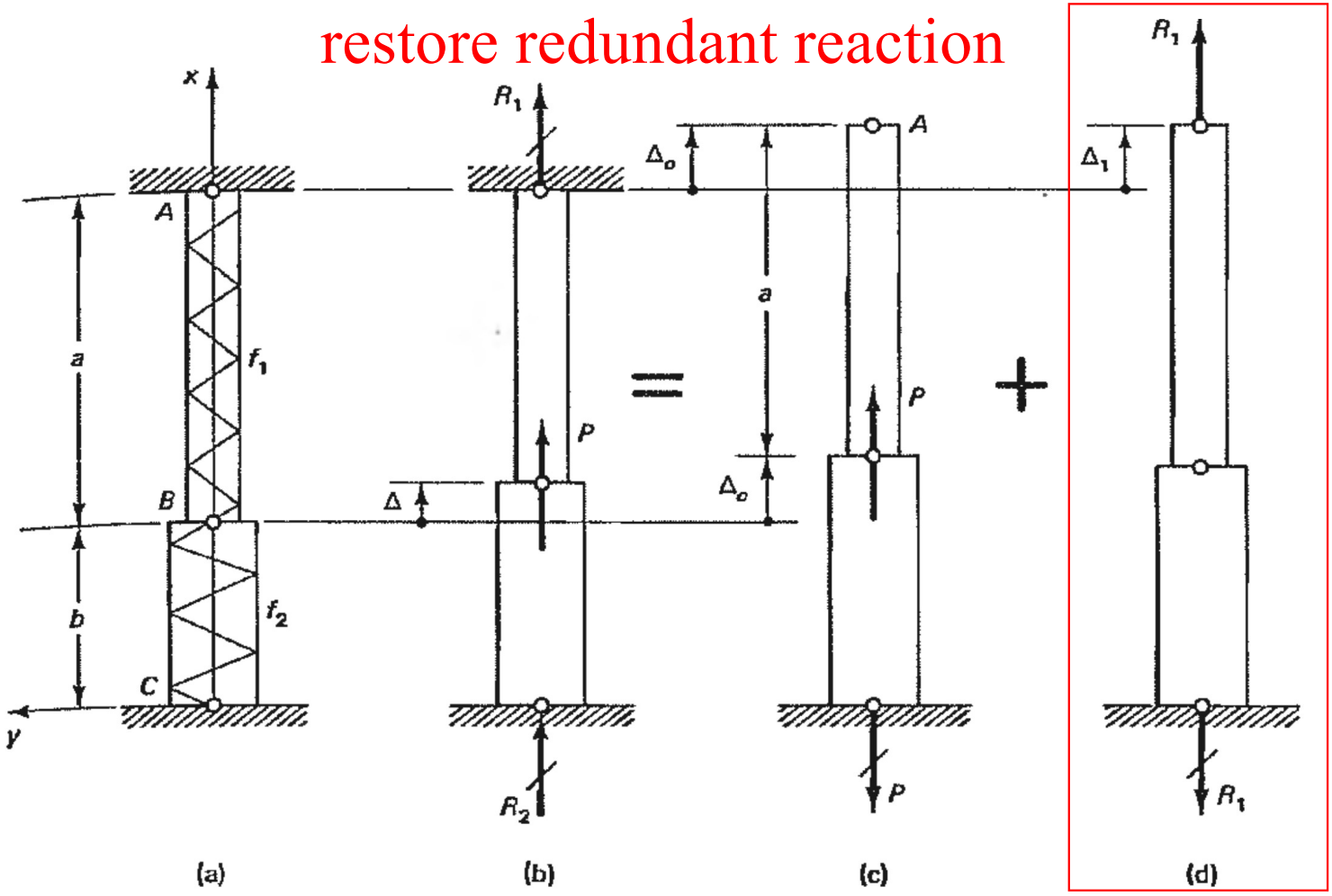
remove redundant reaction



$$\Delta_0 = f_2 P$$

2. Force Method of Analysis - example 1

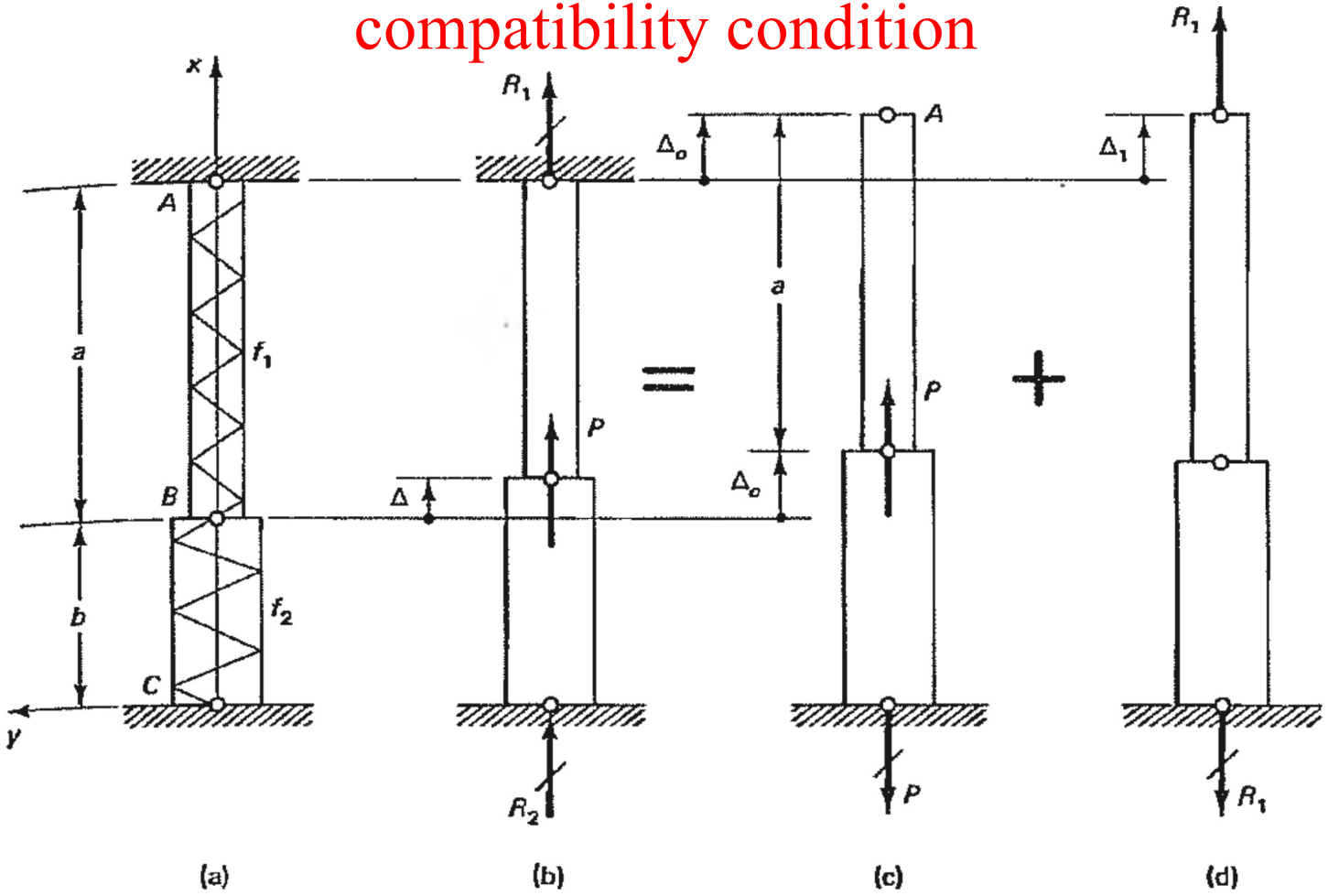
restore redundant reaction



$$\Delta_1 = (f_1 + f_2)R_1$$

2. Force (flexibility) Method of Analysis

compatibility condition



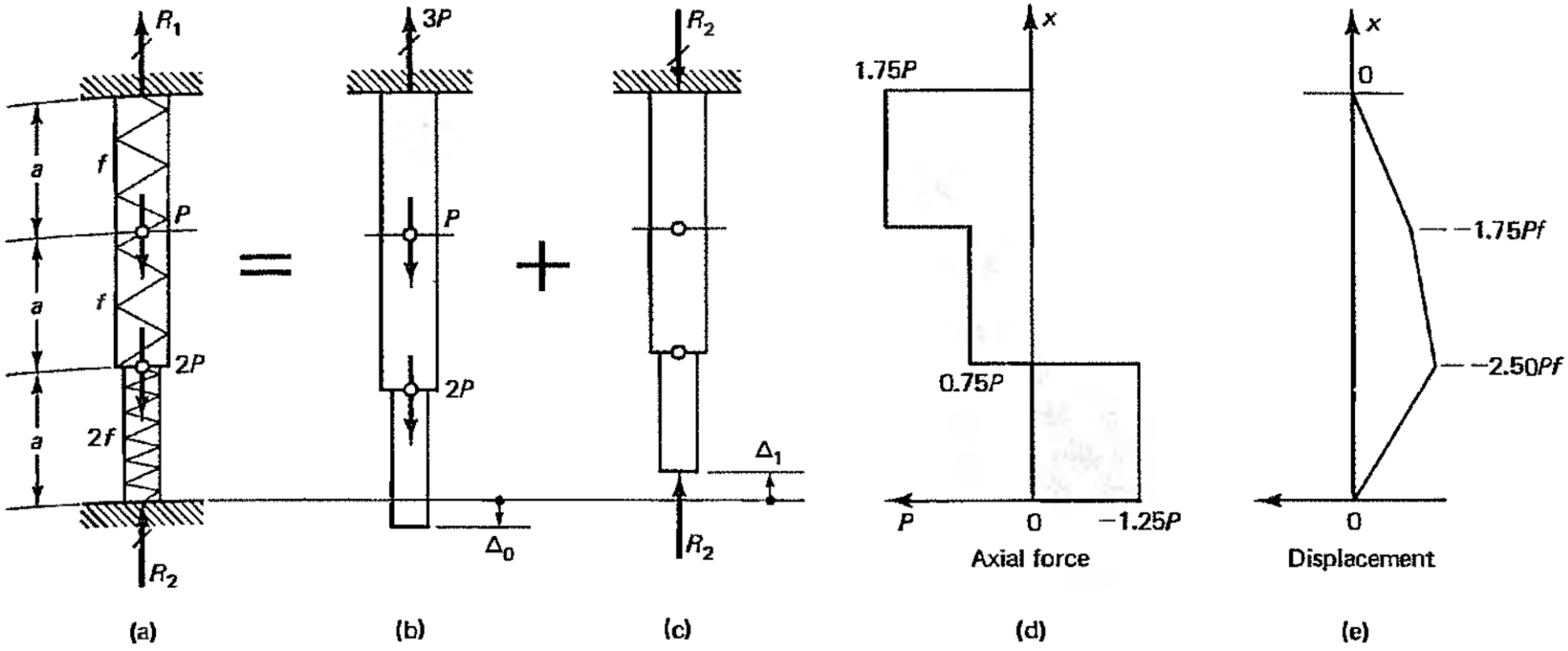
$$\Delta_0 + \Delta_1 = 0$$

$$R_1 = -\frac{f_2}{f_1 + f_2} P$$

$$R_2 = -\frac{f_1}{f_1 + f_2} P$$

2. Force Method of Analysis - example 2

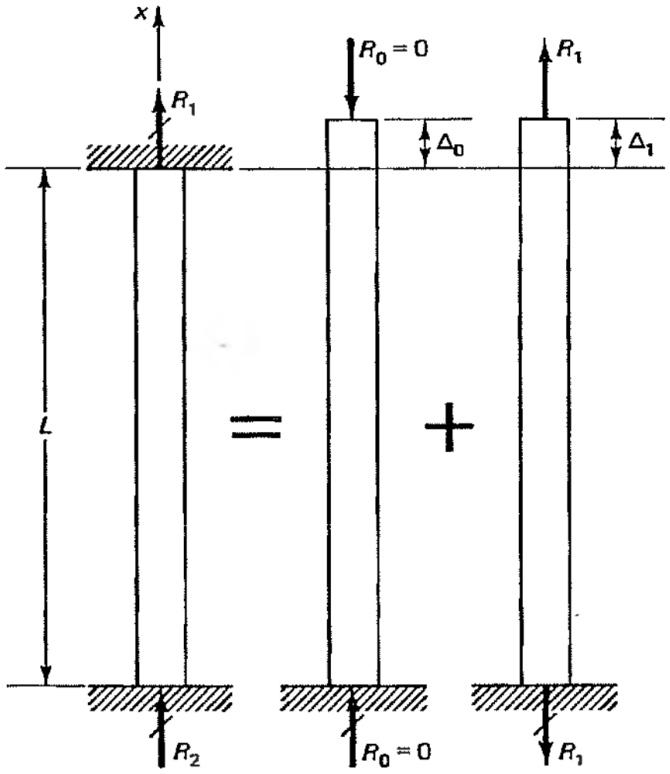
An elastic bar at both ends is loaded as shown in Fig. 2-45. The known flexibility coefficients f and $2f$ for each of the three bar segments are shown in the figure. Determine the reactions and plot the axial force and the axial displacement diagrams for the bar.



2. Force Method of Analysis - example 3

An elastic bar is held at both ends, as shown in Fig. 2-46. If the bar temperature increases by δT, what axial force develops in the bar? AE for the bar is constant.

$$\Delta_0 = \alpha(\delta T)L$$

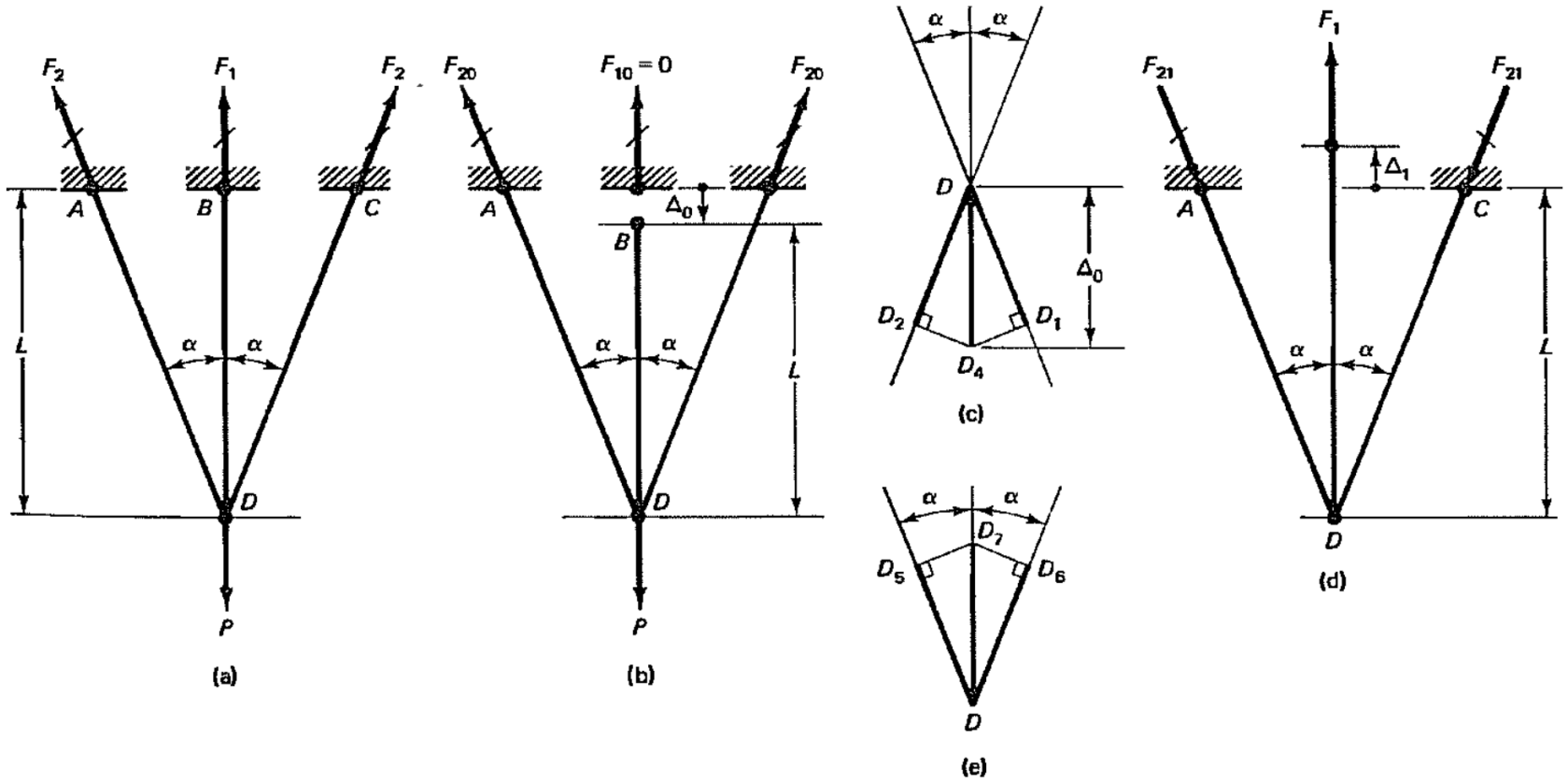


$$\Delta_1 = R_1 f = \frac{R_1 L}{AE}$$

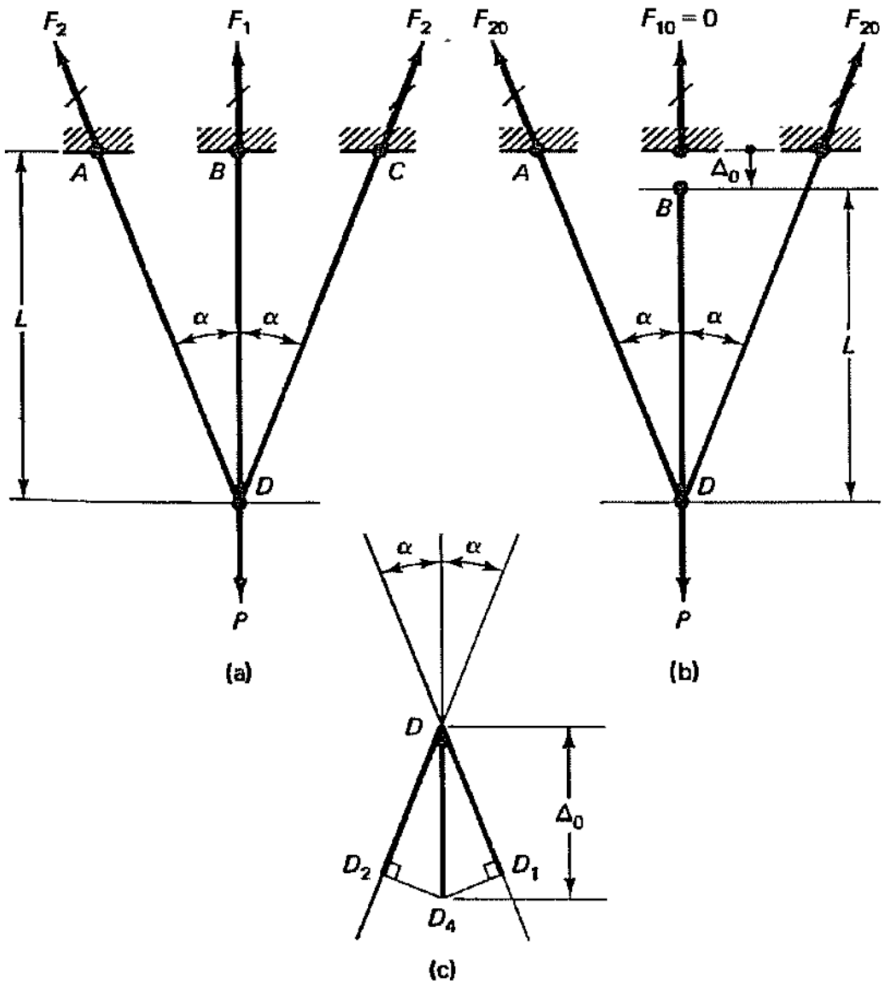
$$\Delta_0 + \Delta_1 = 0, \qquad R_1 = -\alpha(\delta T)AE$$

2. Force Method of Analysis - example 4

For the planar system of the three elastic bars shown in Fig. 2-47(a), determine the forces in the bars caused by applied force P. The cross-sectional area A of each bar is the same, and their elastic modulus is E.



2. Force Method of Analysis - example 4



$$F_{10} = 0 \qquad \text{and} \qquad 2F_{20}\cos\alpha = P$$

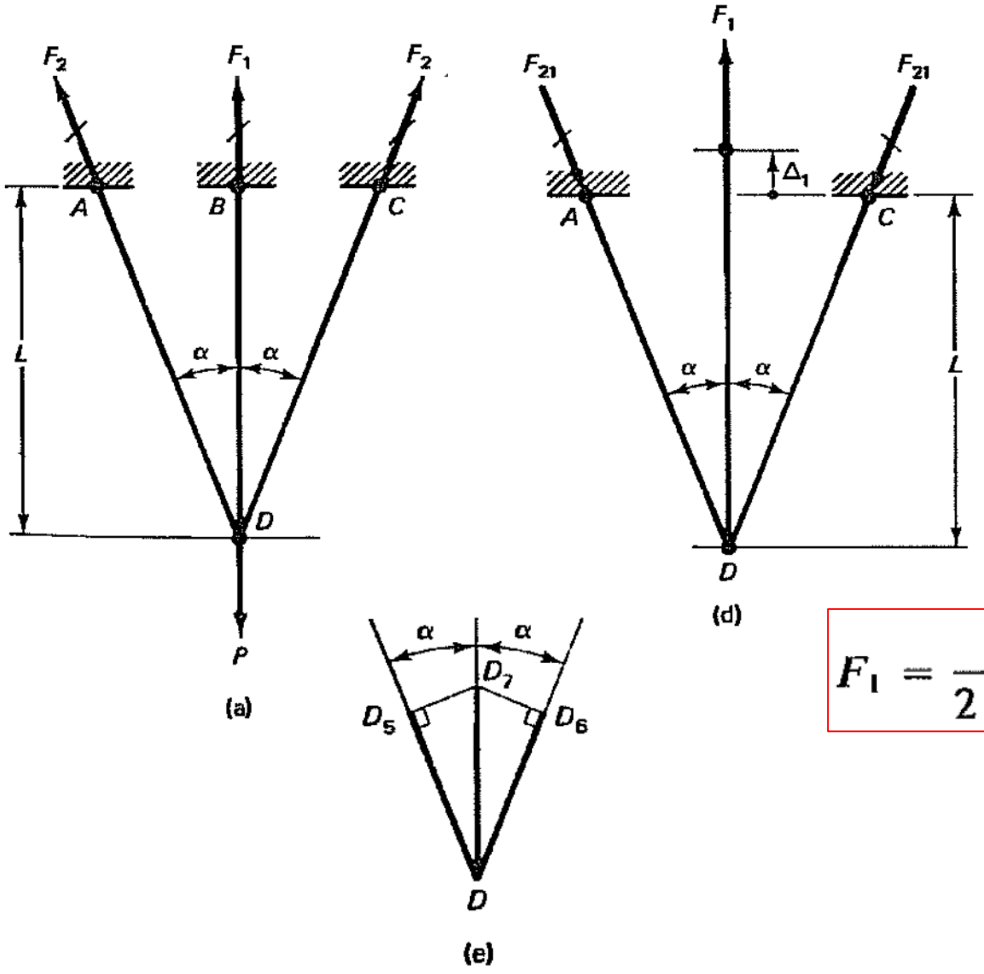
$$F_{20} = \frac{P}{2\cos\alpha}$$

$$L_{AD}\cos\alpha = L, \qquad L_{AD} = L/\cos\alpha$$

$$(\Delta_{AD})_0 = \frac{PL}{2AE\cos^2\alpha}$$

$$\Delta_0\cos\alpha = (\Delta_{AD})_0 \qquad \text{and} \qquad \Delta_0 = -\frac{PL}{2AE\cos^3\alpha}$$

2. Force Method of Analysis - example 4



$$\Delta_1 = \frac{F_1 L}{AE} + \frac{F_1 L}{2AE \cos^3 \alpha}$$

$$\Delta_0 + \Delta_1 = 0,$$

$$F_1 = \frac{P}{2 \cos^3 \alpha + 1} \quad \text{and} \quad F_2 = \frac{P}{2 \cos^3 \alpha + 1} \cos^2 \alpha$$

2. Force Method of Analysis - Comments

The algebraic sum of the two solutions is an application of the principle of superposition. This principle is based upon the premise (前提) that the resultant stress or strain in system due to several forces is the algebraic sum of their effects when separately applied. This assumption is true only if each effect is linearly related to the force causing it.